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THE PROBLEM: SAFE SG → GFL TRANSITION

A replacement is local hardware, but stability is a property of the closed grid and its controller architecture.

SG

grid

→

replace

→

GFL

port i

$\delta_i \in \{0, 1\},$
 $\delta_i = 0$: retain SG _{i}

$S(\delta) = \{i : \delta_i = 1\}$
 $\delta_i = 1$: replace SG _{i} by GFL _{i}

Engineering objective

$\max_{S, K} \sum_{i \in S} P_i^0$ s.t. $\alpha_{\perp}(S, K) \leq -\sigma_{\text{req}}, \quad K \in \mathcal{K}_n$

Report the largest independently verified portfolio found.
Certified global optimality is future work.

A safe SG → GFL transition is a portfolio–controller co-design problem.

P_i^0 : dispatched power of SG _{i} $\alpha_{\perp}$: rightmost physical (non-gauge) root σ_{req} : required decay margin $\mathcal{K}_n$: PLL gains that may use measurements up to n network hops away

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MODEL: DAE → TRANSVERSE STABILITY

3

IEEE-39: MINIMAL COLLECTIVE BLOCKER

4

EXACT NETWORK CLOSURE: LOCAL × COLLECTIVE

5

BEYOND NODAL DAMPING: EFFECTIVE OPERATORS

6

GRID-EMBEDDED PLL GAIN SYNTHESIS

7

INFORMATION RADIUS AND N-HOP CONTROL

8

N-HOP RESULTS: AUTHORITY AND REPLACEMENT

9

DELAYS, CO-DESIGN, AND NONLINEAR CHECKS

MAIN CONTRIBUTIONS

SCOPE OF THE RESULTS

REFERENCES / CONTACT